

Module 2

Limits of functions and continuity

2.1 The concept of limit

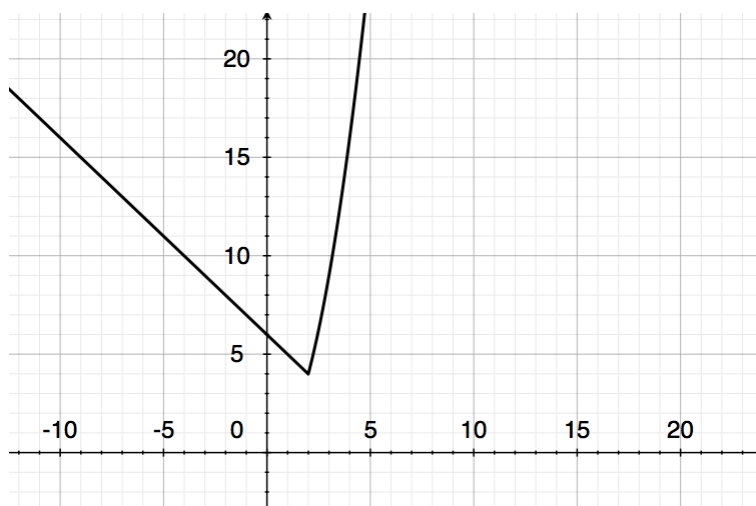
Question (1)

- (a) 1 (b) does not exist (c) 1
(d) $\frac{1}{2}$ (e) 3 (f) 2
(g) $-\infty$ (h) does not exist (i) does not exist

(b) does not exist because the left and right limits are different at $x = 0$; (h) does not exist because $f(x)$ goes to infinity at $x = 4$; (i) does not exist because left and right limit are different at $x = 4$ (i.e. $-\infty, \infty$).

Question (2)

- (a)

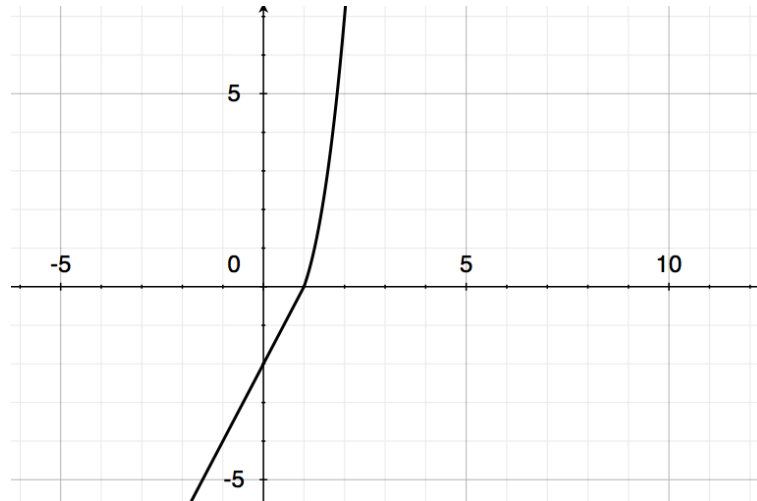


(b)

(a) 4 (b) 4 (c) 4 (d) 4.

Question (3)

(a)



(b)

(a) 0 (b) 0.

2.2 Limit laws**Question (1)**

(a) The solution is given by:

$$\begin{aligned}
 \lim_{x \rightarrow -1} (2x^4 - 3x^3 + 6x + 1) &= \lim_{x \rightarrow -1} 2x^4 - \lim_{x \rightarrow -1} 3x^3 + \lim_{x \rightarrow -1} 6x + \lim_{x \rightarrow -1} 1 && [\text{LAW 1}] \\
 &= 2 \lim_{x \rightarrow -1} x^4 - 3 \lim_{x \rightarrow -1} x^3 + 6 \lim_{x \rightarrow -1} x + \lim_{x \rightarrow -1} 1 && [\text{LAW 2}] \\
 &= 2 \left(\lim_{x \rightarrow -1} x \right)^4 - 3 \left(\lim_{x \rightarrow -1} x \right)^3 + 6 \lim_{x \rightarrow -1} x + \lim_{x \rightarrow -1} 1 && [\text{LAW 3}] \\
 &= 2(-1)^4 - 3(-1)^3 + 6(-1) + 1 && [\text{LAWS 5, 6}] \\
 &= 2 + 3 - 6 + 1 = 0.
 \end{aligned}$$

(b) The solution is now given by:

$$\begin{aligned}
 \lim_{x \rightarrow 1} \frac{2x^2 + 3}{x - 2} &= \frac{\lim_{x \rightarrow 1} (2x^2 + 3)}{\lim_{x \rightarrow 1} (x - 2)} && [\text{LAW 4}] \\
 &= \frac{\lim_{x \rightarrow 1} 2x^2 + \lim_{x \rightarrow 1} 3}{\lim_{x \rightarrow 1} x - \lim_{x \rightarrow 1} 2} && [\text{LAW 1}] \\
 &= \frac{2 \lim_{x \rightarrow 1} x^2 + \lim_{x \rightarrow 1} 3}{\lim_{x \rightarrow 1} x - \lim_{x \rightarrow 1} 2} && [\text{LAW 2}] \\
 &= \frac{2 \left(\lim_{x \rightarrow 1} x \right)^2 + 3}{1 - 2} && [\text{LAW 3, 5, 6}] \\
 &= \frac{2(1)^2 + 3}{-1} && [\text{LAW 5}] \\
 &= -5.
 \end{aligned}$$

(c) The solution is given by:

$$\begin{aligned}
 \lim_{x \rightarrow 2} 2(x^2 - 6)^2 &= \lim_{x \rightarrow 2} 2 \cdot \lim_{x \rightarrow 2} (x^2 - 6)^2 && [\text{LAW 3}] \\
 &= 2 \lim_{x \rightarrow 2} (x^2 - 6) \cdot \lim_{x \rightarrow 2} (x^2 - 6) && [\text{LAW 2, 3}] \\
 &= 2 \left(\left(\lim_{x \rightarrow 2} x \right)^2 - 6 \right) \cdot \left(\left(\lim_{x \rightarrow 2} x \right)^2 - 6 \right) && [\text{LAW 1, 3, 6}] \\
 &= 2(2^2 - 6)(2^2 - 6) && [\text{LAW 5}] \\
 &= 2 \cdot (-2)(-2) = 8.
 \end{aligned}$$

(d) The procedure is similar as that illustrated in the previous solutions. The final result is:

$$\lim_{x \rightarrow 1} \left(\frac{1 - 2x}{2x^2 - x + 1} \right)^4 = \left(\frac{-1}{2} \right)^4 = \frac{1}{16}$$

2.3 Limits at infinity and infinite limits

Question (1)

(a) The solution is given by:

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} (2x^4 - x + 1) &= 2 \left(\lim_{x \rightarrow -\infty} x \right)^4 - \left(\lim_{x \rightarrow -\infty} x \right) + 1 && [\text{LAW 1, 2, 3, 5}] \\
 &= \infty && [\text{LAW 6}].
 \end{aligned}$$

This limit does not give rise to an indefinite form.

(b) We have:

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{2x^2 + x - 3}{x + 7} &= \frac{2 \left(\lim_{x \rightarrow \infty} x \right)^2 + \left(\lim_{x \rightarrow \infty} x \right) - 3}{\left(\lim_{x \rightarrow \infty} x \right) + 7} && [\text{LAW 1, 2, 3, 4, 5}] \\ &= \left(\frac{\infty}{\infty} \right) && [\text{LAW 6}]\end{aligned}$$

(c) The limit exists and the result is $-\infty$.

(d) The limit exists and the result is ∞ .

(e) In this example, if we use limit laws we have two indefinite forms appearing at the same time. In fact:

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^2 - 3x + 6}{x - 10} &= \frac{\left(\lim_{x \rightarrow \infty} x \right)^2 - 3 \left(\lim_{x \rightarrow \infty} x \right) + 6}{\left(\lim_{x \rightarrow \infty} x \right) - 10} && [\text{LAW 1, 2, 3, 4, 5}] \\ &= \left(\frac{(\infty - \infty)}{\infty} \right) && [\text{LAW 6}]\end{aligned}$$

(f) The limit exists and the result is ∞ .

2.4 Simple cases for indefinite forms $\left(\frac{0}{0}\right)$, $\left(\frac{\infty}{\infty}\right)$ and $(\infty - \infty)$

Question (1)

- | | |
|--------------------|--------------------|
| (a) -7 | (b) $\frac{7}{3}$ |
| (c) $\frac{1}{3}$ | (d) $\frac{1}{20}$ |
| (e) 3 | (f) 0 |
| (g) 0 | (h) $-\frac{1}{2}$ |
| (i) $-\frac{1}{4}$ | (j) $\frac{1}{2}$ |

All questions involve standard rationalisation and/or factorisation. In some cases, some

extra consideration has to be taken. A good example of this is question (h). We have:

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 3} - \sqrt{x^2 + x + 1} \right) &= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 - 3} - \sqrt{x^2 + x + 1} \right) \cdot \frac{\left(\sqrt{x^2 - 3} + \sqrt{x^2 + x + 1} \right)}{\left(\sqrt{x^2 - 3} + \sqrt{x^2 + x + 1} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{x^{\cancel{2}} - 3 - x^{\cancel{2}} - x - 1}{\left(\sqrt{x^2 - 3} + \sqrt{x^2 + x + 1} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{-x - 4}{\left(\sqrt{x^2 - 3} + \sqrt{x^2 + x + 1} \right)} \\
 &= \lim_{x \rightarrow \infty} \frac{x \left(-1 - \frac{4}{x} \right)}{x \left(\sqrt{1 - \frac{3}{x}} + \sqrt{1 + \frac{1}{x} + \frac{1}{x^2}} \right)} = \frac{-1 - 0}{1 + 1} = -\frac{1}{2},
 \end{aligned}$$

since terms $1/x$ and $1/x^2$ go to zero for $x \rightarrow \infty$. Note that, in rationalising the first square root term in the denominator, we used the following:

$$\sqrt{x^2 - 3} = \sqrt{x^2 \left(1 - \frac{3}{x^2} \right)} = \sqrt{x^2} \cdot \sqrt{\left(1 - \frac{3}{x^2} \right)} = |x| \sqrt{\left(1 - \frac{3}{x^2} \right)}.$$

Given that we are concerned with a limit for $x \rightarrow \infty$, only large, positive x are involved. Hence we can assume safely that $x > 0$ and so $|x| = x$. This implies:

$$\sqrt{x^2 - 3} = x \sqrt{\left(1 - \frac{3}{x^2} \right)}.$$

A similar procedure is applied to the other term in the denominator, i.e. $\sqrt{x^2 + x + 1}$.

2.5 The sandwich theorem and standard limits

Question (1) All the limits here evaluate to 0.

(a) Use the fact that: $-1 \leq \sin\left(\frac{7\pi}{x}\right) \leq 1$. Be careful when multiplying by x^3 , since the limit is for $x \rightarrow 0$ and both cases $x > 0$ and $x < 0$ are relevant. You can observe that the inequalities change their signs when $x < 0$, swapping the sandwiching functions together, but this does not affect the final result.

(b) Use the fact that: $0 \leq \cos^2 x \leq 1$. Note that $e^{-x} > 0$, always.

(c) Use the fact that $-\frac{\pi}{2} < \arctan(3x) < \frac{\pi}{2}$. Explicitly indicate that a division by $\sqrt{x}$ is not problematic, since $x \rightarrow \infty$ and so $x > 0$.

(d) Use the fact that: $-1 \leq \sin\left(\frac{x+3}{x-2}\right) \leq 1$. Be sure to indicate that, since we are under the limit operation, $x \rightarrow 2$ and so $x \neq 2$.

Question (2)

- | | |
|---------------------------|--------------------|
| (a) 2 | (b) 0 |
| (c) $-\frac{1}{2}$ | (d) $\frac{2}{5}$ |
| (e) $\frac{1}{3}$ | (f) $\frac{1}{7}$ |
| (g) $\frac{1}{2}$ | (h) $\frac{1}{2}$ |
| (i) $\frac{1}{4\sqrt{2}}$ | (j) $\frac{1}{12}$ |

Let us see some explicitly worked out solutions. In all these examples, the expression given needs to be slightly reworked using normal algebraic processes, so that a standard limit is evident. For question (c), we have:

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\cos x - \sin^2 x - 1}{3x^2} &= \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} - \lim_{x \rightarrow 0} \frac{\sin^2 x}{3x^2} && \text{[Using limit laws]} \\
 &= -\frac{1}{3} \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \right) - \frac{1}{3} \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 && \text{[Using limit laws]} \\
 &= -\frac{1}{3} \left(\frac{1}{2} \right) - \frac{1}{3} (1)^2 && \text{[Using SL. 1 and SL. 2]} \\
 &= -\frac{1}{6} - \frac{1}{3} = -\frac{1}{2}
 \end{aligned}$$

For question (d):

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{5x} &= \lim_{x \rightarrow 0} \frac{(e^x - 1)(e^x + 1)}{5x} = \frac{1}{5} \left(\lim_{x \rightarrow 0} (e^x + 1) \right) \cdot \left(\lim_{x \rightarrow 0} \frac{e^x - 1}{x} \right) && \text{[Using limit laws]} \\
 &= 2 \cdot \frac{1}{5} \cdot 1 = \frac{2}{5},
 \end{aligned}$$

where we used SL. 3 in the last line.

For question (f):

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\tan x}{7x} &= \frac{1}{7} \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x} = \frac{1}{7} \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x} && \text{[Using limit laws]} \\
 &= \frac{1}{7} \cdot 1 \cdot 1 = \frac{1}{7},
 \end{aligned}$$

where we used SL. 1 in the last line.

For question (h):

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\ln(1 - x^2) - \ln(1 - x)}{2x} &= \lim_{x \rightarrow 0} \frac{\ln \left(\frac{1 - x^2}{1 - x} \right)}{2x} = \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{2x} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = \frac{1}{2} \cdot 1 = \frac{1}{2}.
 \end{aligned}$$

where we used limit laws and SL. 5 in the last line. We also simplified inside the logarithm, using the identity $(1 - x^2) = (1 + x)(1 - x)$.

For asterisked question (i*): first use rationalisation, then use SL. 4. For (j*), rationalisation works straightforwardly, otherwise note that

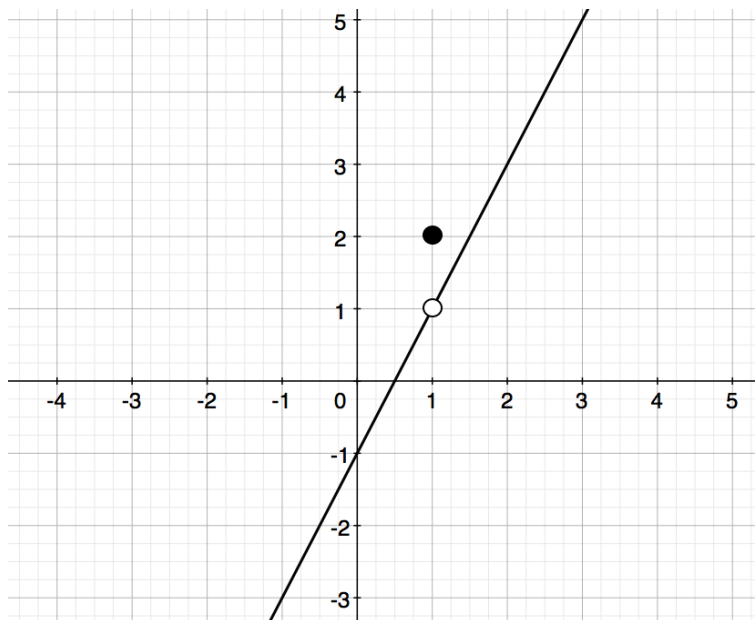
$$(\sqrt{1+x} - 1)^2 = (1+x) + 1 - 2\sqrt{1+x} = 2+x - 2\sqrt{1+x}$$

and use SL. 4.

2.6 Continuity: definition and examples

Question (1)

(a)



(b)

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (2x - 1) = 1$$

(c)

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x - 1) = 1$$

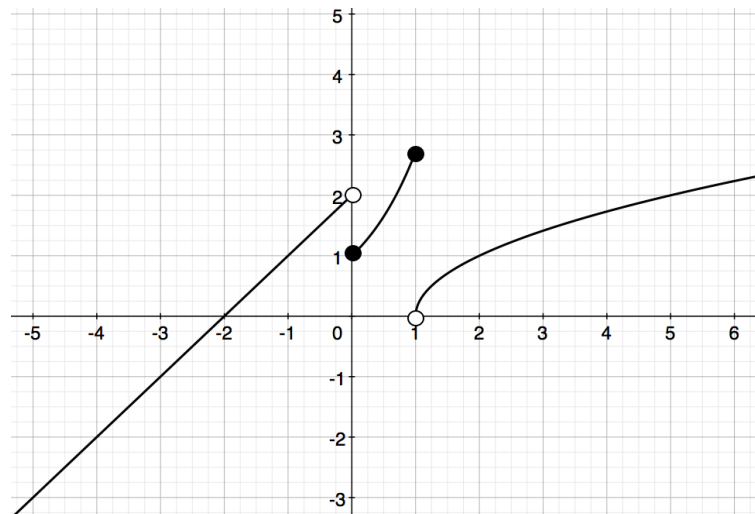
(d) Not continuous, since

$$\lim_{x \rightarrow 1} f(x) = 1 \neq f(1) = 2.$$

The discontinuity is a removable discontinuity: the function has a hole.

Question (2)

(a)



(b) We have:

$$\begin{aligned}\lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} e^x = e \\ \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} \sqrt{x-1} = 0\end{aligned}$$

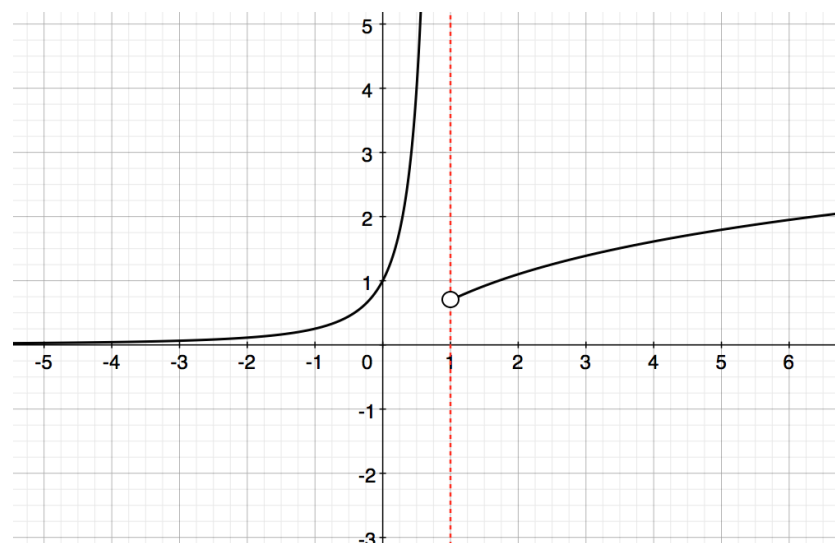
(c) We have:

$$\begin{aligned}\lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (x+2) = 2 \\ \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} e^x = 1\end{aligned}$$

(d) At $x = 0$ and $x = 1$ the function has jump discontinuities, because the left and right limits exist and are finite but different.

Question (3)

(a)



(b) Given that:

$$\begin{aligned}\lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} \frac{1}{(x-1)^2} = \infty \\ \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} \ln(x+1) = \ln 2,\end{aligned}$$

the discontinuity is essential.

2.7 Theorems on continuity and applications to limits

Question (1)

(a) The given function is

$$y = x^2 + \sqrt{3x-2},$$

so

- x^2 is a polynomial, continuous for every $x \in \mathbb{R}$.
- $\sqrt{3x-2}$ is a square root, continuous in its natural domain D (i.e. for positive values of the argument), where

$$D = [2/3, \infty).$$

y is a sum of two continuous functions, hence it is continuous in $D = [2/3, \infty)$.

In this and some subsequent questions, there is a subtlety that you need to be aware of. You have been taught that for continuity, we need the limits to be identical for approach from either side of the point in question, and to equal the value of the function at that point. It might not have been emphasised that this condition applies only in the domain considered and does not need to apply outside of that domain. For the square-root function $f(x) = \sqrt{x}$, values $x < 0$ are not in the domain and so we do not need to look at the limit from that side. We **do** still have equality of the right-sided limit and the function value at $x = 0$ and that is enough for continuity at 0 of this function.

(b) The given function is

$$y = \frac{2x+7}{2x^2-x+1},$$

so

- $(2x+7)$ is a polynomial, continuous for every $x \in \mathbb{R}$.
- $(2x^2-x+1)$ is a polynomial, continuous for every $x \in \mathbb{R}$.

y is a rational function, with both numerator and denominator continuous everywhere. It is thus, overall, continuous in its natural domain, given by all x such that

$$2x^2 - x + 1 \neq 0.$$

Given that the discriminant of the quadratic above is always negative, i.e.

$$\Delta = 1^2 - 4 \cdot 2 \cdot 1 = -7 < 0,$$

we conclude that y is continuous for every $x \in \mathbb{R}$ because the denominator cannot be zero for any real value of x .

(c) The given function is

$$y = \sqrt{2 + |x|} - \sqrt{2 - |x|},$$

so

- $\sqrt{2 + |x|}$ is a square root, continuous in its natural domain, hence for those x such that

$$2 + |x| \geq 0 \quad \Rightarrow \quad |x| \geq -2.$$

The absolute value function has the property $|x| \geq 0$ for every $x \in \mathbb{R}$, hence $\sqrt{2 + |x|}$ is continuous everywhere in $\mathbb{R}$.

- $\sqrt{2 - |x|}$ is also a square root, continuous in its natural domain, hence for those x such that

$$2 - |x| \geq 0 \quad \Rightarrow \quad |x| \leq 2.$$

This is true for $x \in [-2, 2]$, hence $\sqrt{2 - |x|}$ is continuous in $D = [-2, 2]$.

y is a difference of two continuous functions with different domains of continuity and is therefore continuous in the smaller of those domains, namely $D = [-2, 2]$.

(d) The given function is

$$y = \ln \left(\frac{x^2 - 2}{\sqrt{x - 1} - 1} \right).$$

It is a logarithm, so it is continuous in its natural domain, i.e. for those values x such that

$$\frac{x^2 - 2}{\sqrt{x - 1} - 1} > 0.$$

The fraction contains the square root $\sqrt{x - 1}$, which is continuous in its natural domain $D = [1, \infty)$.

To establish the domain of continuity for $\ln$, we have to consider the following two inequalities

$$\begin{array}{lll} x^2 - 2 > 0 & \Rightarrow & x > \sqrt{2} \text{ or } x < -\sqrt{2} \\ \sqrt{x - 1} - 1 > 0 & \Rightarrow & x > 2 \end{array}$$

We can draw up a table of signs:

	$x < -\sqrt{2}$	$-\sqrt{2} < x < \sqrt{2}$	$\sqrt{2} < x < 2$	$x > 2$
$x^2 - 2 > 0$	+ve	-ve	+ve	+ve
$\sqrt{x-1} - 1 > 0$	-ve	-ve	-ve	+ve
$\frac{x^2 - 2}{\sqrt{x-1} - 1}$	-ve	+ve	-ve	+ve

Using this result and remembering that only solutions in $D = [1, \infty)$ are acceptable (i.e. the domain of continuity for $\sqrt{x-1}$), the function y is continuous in

$$A = [1, \sqrt{2}) \cup (2, \infty).$$

(e) y is continuous in $A = (-1, 0) \cup (0, 1)$.

(f) y is continuous in $A = \mathbb{R} \setminus \{1, 2\}$.

(g) y is continuous in $A = (0, 2)$.

(h) y is continuous in $A = [-1, 1)$.

Question (2)

In all the previous examples, the given functions are continuous on their natural domain. This is because they are obtained using products, ratios or sums/differences of standard continuous functions, such as logarithms, square roots, etc.

Question (3)

$$k = 2.$$

Question (4)

$$k = -\frac{2}{3}, p = -\frac{1}{3}.$$

Question (5)

- (a) $\ln 2$ (b) 0
- (c) $\frac{1}{e}$ (d) D.N.E. The limit is $\pm\infty$ depending on which side of 1 we approach from.
- (e) 1 (f) $(\sqrt{2} - 4)^{1/3}$
- (g) $\arctan(-\cosh(1))$ (or, using parity, $-\arctan(\cosh(1))$)

Let us work through some of these.

Question (a):

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\ln(x + \sqrt{x+4})}{\cos(2\pi x^4)} &= \frac{\ln\left(\lim_{x \rightarrow 0} (x + \sqrt{x+4})\right)}{\cos\left(2\pi \lim_{x \rightarrow 0} x^4\right)} && \text{limit laws, continuity of } \ln z \text{ for } z > 0 \\
 &&& \text{and continuity of } \cos z \text{ for every } z \in \mathbb{R} \\
 &= \frac{\ln\left(\lim_{x \rightarrow 0} x + \sqrt{\lim_{x \rightarrow 0} x + 4}\right)}{\cos\left(2\pi \left(\lim_{x \rightarrow 0} x\right)^4\right)} && \text{limit laws, continuity of } \sqrt{z} \text{ for } z \geq 0 \\
 &= \frac{\ln(0 + \sqrt{0+4})}{\cos(2\pi \cdot 0)} = \ln 2,
 \end{aligned}$$

where limit laws have been used in the last line.

Question (b):

$$\begin{aligned}
 \lim_{x \rightarrow 0} \sin(\sin x + \pi) &= \sin\left(\lim_{x \rightarrow 0} (\sin x + \pi)\right) && \text{continuity of } \sin z \text{ for every } z \in \mathbb{R} \\
 &= \sin\left(\sin\left(\lim_{x \rightarrow 0} x\right) + \pi\right) && \text{limit laws, continuity of } \sin z \text{ for every } z \in \mathbb{R} \\
 &= \sin(0 + \pi) = 0.
 \end{aligned}$$

Question (c):

$$\begin{aligned}
 \lim_{x \rightarrow 1} \exp\left(\frac{2x-1}{x^2+4x-6}\right) &= \exp\left(\lim_{x \rightarrow 1} \frac{2x-1}{x^2+4x-6}\right) && \text{continuity of } e^z \text{ for every } z \in \mathbb{R} \\
 &= \exp\left(\frac{2 \cdot 1 - 1}{1^2 + 4 \cdot 1 - 6}\right) = e^{-1} = \frac{1}{e},
 \end{aligned}$$

where limit laws have been used in the last line.

The other questions can be completed using the following hints:

Question (d): Use limit laws and continuity of $\sqrt[3]{z}$ for every $z \in \mathbb{R}$, $\ln z$ for $z > 0$ and $\arcsin z$ for $z \in [-1, 1]$.

Question (e): Consider that $|z| \geq 0$ for every $z \in \mathbb{R}$. Use limit laws and continuity of z^z for every $z \geq 0$. Note that the equal sign in the previous inequality is a result of SL. 7.

Question (f): Use limit laws and continuity of $\sqrt[3]{z}$ for $z \in \mathbb{R}$, $\sqrt{z}$ for $z \geq 0$.

Question (g): Use limit laws and continuity of $\sinh z$, $\cosh z$ for every $z \in \mathbb{R}$, continuity of $\sqrt{z}$ for $z \geq 0$ and continuity for $\arctan z$ for $z \in \mathbb{R}$.

Note: In this question a clarification along the lines of that after Question 2.7. 1(a) is appropriate again. The limit of $\sqrt{x}$ exists at $x = 0$ despite the inability to approach from the left. We only need to operate within the domain and not outside it for the limit to exist.

2.8 The “race to infinity”: comparing “speeds” for the $\left(\frac{\infty}{\infty}\right)$ case

Question (1)

- (a) ∞ ; x^{13} (num) is faster than x^{11} (den).
- (b) 0; e^x (den) is faster than x^{20} (num).
- (c) 0; $\sin x$, $\cos x$ (num) do not “go to infinity”, since they oscillate and their ranges are bounded between $[-1, 1]$. x^2 (den) is the fastest in “going to infinity”.
- (d) The limit does not exist; $\tanh x \rightarrow 1$ as $x \rightarrow \infty$ (den), whereas $\tan x$ is periodic and does not have a limit when $x \rightarrow \infty$.
- (e) ∞ ; x^{2x} (num) is faster than $(2x)^x$ (den). In fact, using the properties of powers we can write:

$$(2x)^x = 2^x \cdot x^x,$$

and the limit is then evaluated as follows:

$$\lim_{x \rightarrow \infty} \frac{x^{2x}}{(2x)^x} = \lim_{x \rightarrow \infty} \frac{x^{\cancel{2x}}}{2^x \cancel{x^x}} = \lim_{x \rightarrow \infty} \frac{x^x}{2^x} = \infty,$$

since x^x “goes to infinity” faster than any exponential.

- (f) 0. Note that $\ln(1 + e^x)$ (num) behaves almost identically to $\ln(e^x)$ as for large x , 1 is negligible in comparison to the exponential. Hence the numerator behaves as $\ln(e^x) = x$. The denominator grows faster i.e. x^2 (den) grows faster than x (num), so the limit is 0.

2.9 The substitution technique and further standard limits

Question (1)

- (a) 0
- (b) 1
- (c) $1 + e$
- (d) e^3
- (e) e^4
- (f) 1

Let us consider some worked out examples. Question (a):

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} (x^2 \ln x - x \ln(x^2)) &= \lim_{x \rightarrow 0^+} (x^2 \ln x - 2x \ln x) \\
 &= \left(\lim_{x \rightarrow 0^+} x \right) \cdot \underbrace{\left(\lim_{x \rightarrow 0^+} x \ln x \right)}_{\text{SL. 6}} - 2 \underbrace{\left(\lim_{x \rightarrow 0^+} x \ln x \right)}_{\text{SL. 6}} \quad \text{using limit laws} \\
 &= 0 \cdot 0 - 2 \cdot 0 = 0,
 \end{aligned}$$

where limit law 6 has been used in the last line.

Question (b) can be solved using limit laws and the fact that $(2x)^x = 2^x \cdot x^x$, i.e.

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} (x^x + (2x)^x - x^{-x}) &= \lim_{x \rightarrow 0^+} x^x + \lim_{x \rightarrow 0^+} (2x)^x - \lim_{x \rightarrow 0^+} x^{-x} \\
 &= \underbrace{\lim_{x \rightarrow 0^+} x^x}_{\text{SL. 7}} + \lim_{x \rightarrow 0^+} 2^x \cdot \underbrace{\lim_{x \rightarrow 0^+} x^x}_{\text{SL. 7}} - \underbrace{\lim_{x \rightarrow 0^+} x^{-x}}_{\text{SL. 7}} \\
 &= 1 + 1 \cdot 1 - 1 = 2 - 1 = 1.
 \end{aligned}$$

For question (c), we use the fact that

$$e^{1-x \ln x} = e \cdot e^{-x \ln x} = ex^{-x}$$

and that

$$x^2 \ln^2 x + 2x \ln x + 1 = (x \ln x + 1)^2.$$

Then, SL 6 and SL. 7 should be used.

For question (d), we have to remember that

$$\left(1 + \frac{1}{x}\right)^{3x} = \left[\left(1 + \frac{1}{x}\right)^x\right]^3$$

and use SL. 8.

For question (e), use

$$1 + x^2 + 2x = (1 + x)^2.$$

Then, SL. 8 gives the final result.

For question (f), use continuity of 2^z and 3^z for $z \in \mathbb{R} > 0$. Then use SL. 7. Remember that properties of powers imply that $(3^x)^x = 3^{x^2}$.

Question (2)

Please note that these exercises are intended as practice on the substitution technique. Some of the processes are lengthy and in practice there might sometimes be quicker, more efficient routes to the limit.

- | | |
|--------------------|--------------------|
| (a) e^2 | (b) $-\frac{1}{2}$ |
| (c*) $\frac{1}{2}$ | (d) 1 |
| (e*) 1 | (f*) 1 |
| (g) 1 | (h*) $\frac{1}{2}$ |

Some hints and worked out solutions are below.

In question (a), use SL. 8.

In question (b), use SL. 2. Be careful with the overall sign of the expression.

For question (c*), we can use the definition of $\tan x = \frac{\sin x}{\cos x}$ to find an expression for $\sin x$ in terms of the tangent:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - 1}{\sin x} = \lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - 1}{\tan x \cos x}.$$

At this point, to be able to perform the substitution, we have to recast $\cos x$ in terms of $\tan x$. Using one of the fundamental identities of trigonometric functions, we have:

$$\sec^2 x = 1 + \tan^2 x \quad \Rightarrow \quad \frac{1}{\cos^2 x} = 1 + \tan^2 x \quad \Rightarrow \quad \cos x = \pm \frac{1}{\sqrt{1 + \tan^2 x}}.$$

Since the limit is for $x \rightarrow 0$, we are considering values x in the vicinity of zero. For both values $x > 0$ and $x < 0$, we have that $\cos x > 0$, because x is in the first or fourth quadrant, respectively. So we can choose the positive sign and substitute inside the limit as follows:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - 1}{\tan x \cdot \frac{1}{\sqrt{1 + \tan^2 x}}} = \lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - 1}{\tan x} \cdot \sqrt{1 + \tan^2 x}.$$

If we perform the given substitution $t = \tan x$, observing that $t \rightarrow 0$ when $x \rightarrow 0$ since $\tan x \rightarrow 0$, we find:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan x} - 1}{\tan x} \cdot \sqrt{1 + \tan^2 x} = \lim_{t \rightarrow 0} \frac{\sqrt{1 + t} - 1}{t} \cdot \sqrt{1 + t^2} = \underbrace{\lim_{t \rightarrow 0} \frac{\sqrt{1 + t} - 1}{t}}_{\text{SL. 4}} \cdot \lim_{t \rightarrow 0} \sqrt{1 + t^2},$$

where we have used the limit law for products in the last line. It is now straightforward, using SL. 4 and the continuity of $\sqrt{z}$ for $z \geq 0$, to find that the final result is $1/2$.

For question (d), use limit laws and SL. 1. If $t = \arcsin x$, then $x = \sin t$.

Question (e*) can be solved as follows:

$$\begin{aligned}\lim_{x \rightarrow \infty} \left(1 + \frac{1}{\sinh x}\right)^{\tanh x} &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\sinh x}\right)^{\frac{\sinh x}{\cosh x}} \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\sinh x}\right)^{\frac{\sinh x}{\sqrt{1+\sinh^2 x}}},\end{aligned}$$

where we used the fundamental identity of hyperbolic function $\cosh^2 x - \sinh^2 x = 1$. Note that $\cosh x$ is positive for large values of positive x (remember that the limit is for $x \rightarrow \infty$) and so we can use

$$\cosh x = \sqrt{1 + \sinh^2 x}.$$

We can now substitute $t = \sinh x$, noting that $t \rightarrow \infty$ because $\sinh x \rightarrow \infty$ when $x \rightarrow \infty$:

$$\begin{aligned}\lim_{x \rightarrow \infty} \left(1 + \frac{1}{\sinh x}\right)^{\frac{\sinh x}{\sqrt{1+\sinh^2 x}}} &= \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^{\frac{t}{\sqrt{1+t^2}}} \\ &= \lim_{t \rightarrow \infty} \exp \left(\frac{\ln \left[\left(1 + \frac{1}{t}\right)^t \right]}{\sqrt{1+t^2}} \right).\end{aligned}$$

In the last step, we used the exponential form, i.e. $y^z = \exp(z \cdot \ln y)$, with

$$\begin{aligned}z &= \frac{1}{\sqrt{1+t^2}} \\ y &= \left(1 + \frac{1}{t}\right)^t.\end{aligned}$$

Using continuity of $\ln z$ for $z > 0$, continuity for $\sqrt{z}$ for $z \geq 0$ and continuity of e^z for every $z \in \mathbb{R}$ (remember that the limit is for $x \rightarrow \infty$, so only very large and positive values are involved), we find:

$$\begin{aligned}\lim_{t \rightarrow \infty} \exp \left(\frac{\ln \left[\left(1 + \frac{1}{t}\right)^t \right]}{\sqrt{1+t^2}} \right) &= \exp \left(\frac{\ln \left[\overbrace{\lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t}^{\text{SL. 8}} \right]}{\sqrt{1 + \lim_{t \rightarrow \infty} t^2}} \right) \\ &= \exp \left(\frac{\ln e}{\sqrt{1 + \lim_{t \rightarrow \infty} t^2}} \right) = \exp(0) = 1,\end{aligned}$$

where we used limit laws in the last passage.

Solution to question (f*) is as follows:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1)^2}{\tan^2 x} &= \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1)^2}{\frac{\sin^2 x}{\cos^2 x}} = \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1)^2}{\sin^2 x} \cdot \cos^2 x \\ &= \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1)^2 (1 - \sin^2 x)}{\sin^2 x},\end{aligned}$$

where the definition of $\tan x$ and the fundamental identity $\cos^2 x = 1 - \sin^2 x$ have been used. We can now perform the substitution, observing that $t \rightarrow 0$ because $\sin x \rightarrow 0$ when $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1)^2 (1 - \sin^2 x)}{\sin^2 x} &= \lim_{t \rightarrow 0} \frac{(e^t - 1)^2 (1 - t^2)}{t^2} \\ &= \lim_{t \rightarrow 0} \frac{(e^t - 1)^2}{t^2} \cdot \lim_{t \rightarrow 0} (1 - t^2) = \lim_{t \rightarrow 0} \left(\frac{e^t - 1}{t} \right)^2 \cdot \lim_{t \rightarrow 0} (1 - t^2) \\ &= \underbrace{\left(\lim_{t \rightarrow 0} \frac{e^t - 1}{t} \right)^2}_{\text{SL. 3}} \cdot \lim_{t \rightarrow 0} (1 - t^2) = 1^2 \cdot 1 = 1,\end{aligned}$$

where limit laws and SL. 3 have been used.

For question (g), use the fundamental identity for hyperbolic functions, i.e. $\cosh^2 x - \sinh^2 x = 1$ and, after some simple simplification, arrive at SL. 7.

Question (h*) is tricky but simple. First, consider the fundamental identity of trigonometric functions and rewrite the given limit as follows

$$\lim_{x \rightarrow 0} \frac{\ln(2 - \cos x)}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{\ln(1 + (1 - \cos x))}{1 - \cos^2 x}.$$

Using the given substitution, we can find an expression for $1 - \cos^2 x$ in terms of t as follows:

$$t = 1 - \cos x \quad \Rightarrow \quad \cos x = 1 - t \quad \Rightarrow \quad \cos^2 x = (1 - t)^2.$$

Given this result, $1 - \cos^2 x$ in terms of t is given by:

$$1 - \cos^2 x = 1 - (1 - t)^2 = 1 - (1 - 2t + t^2) = 2t - t^2 = t(2 - t).$$

We can now substitute, noting that $t \rightarrow 0$ as $\cos x \rightarrow 1$ for $x \rightarrow 0$, and write:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\ln(1 + (1 - \cos x))}{1 - \cos^2 x} &= \lim_{t \rightarrow 0} \frac{\ln(1 + t)}{t(2 - t)} = \lim_{t \rightarrow 0} \underbrace{\frac{\ln(1 + t)}{t}}_{\text{SL. 5}} \cdot \lim_{t \rightarrow 0} \frac{1}{2 - t} \\ &= 1 \cdot \frac{1}{2 - 0} = \frac{1}{2},\end{aligned}$$

where limit laws and SL. 5 have been used.